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## Trajectory generation using predictive PID control for stable walking humanoid robot

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### Abstract

Predictive control presents a solution to the problem of the trajectory generation for a humanoid robot. This control strategy is introduced by Katayama<sup>13</sup> and improved by Kajita<sup>1</sup>, in those literatures, the theory of predictive control is used to calculate the trajectory of the mass center of the cart table to track a reference trajectory and give it for the ZMP. However this control has an inconvenience such as the non respect of real-time constraints, due to the computation time and the output response establishing. In this paper we propose the predictive PID controller that imitate the calculated time and reduce the complexity of the control algorithm. The aim of this new control approach for biped robot is to make walking smoother and more efficient.

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**Keywords:** Walking robot, Cart table model, ZMP model, Predictive PID;

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### 1. Introduction

The control of autonomous robots is a complex problem due to the presence of many parameters. This becomes more complicate for biped humanoid robots because they are naturally unstable when moving and this make the task of control more difficult. Humanoid robots control still need to be greatly improved, both in terms of their modelling and in the design of their control. Contrary to robot manipulators, the mass centre and the notions of the Zero Point Moment are predominant among humanoids and its will be central to the management of their balance.

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This paper fits in this context. The control theory involves physical interaction between an articulated system and its environment. This close relationship is in fact a common set of fundamental problems such as the planning and implementation of robust stable dynamic movements. Those problems are for the most part, far from being resolved.

The most substantial fundamental problem with this physical interaction is the laws that govern the contact forces between the robot and the ground. But many usual methods of the theory of control cannot ensure a satisfying control of such dynamic systems. The solution could be the predictive control methods that consider the dynamic behaviour in its entirety<sup>1,2</sup>. In this context, this paper which aims to provide a new control strategy adapted to humanoid robots whose complexity continues to increase.

In many works in humanoid robotic, the PID controller presents a solution for the problem of the control walk motion of the robot<sup>6</sup>. In this study, we are interested in the predictive PID control to improve the performance of PID controller. We used the predictive characteristic of GPC control to synthesis this proposed controller<sup>5</sup>. The advantage of this proposed method is to simplify the implementation in real time processing. The Predictive PID solves the problem of Optimal Control in a prediction horizon. The main advantage of this proposed method is that the input variables and future trajectories can easily be incorporated into the control design.

The first part of this paper presents the ZMP Preview Control scheme, proposed in<sup>1,9</sup>, in order to generate dynamically stable motions through a cart table approximation for the dynamics of the Mass Centre of a humanoid robot. The second part presents the generalized predictive control GPC algorithm. This algorithm law is used in the design of the proposed controllers. The third part presents the predictive PID control based on GPC control.

## 2. Cart-table model

The cart table model presented in (figure 1) is proposed by<sup>1</sup> in objective to simplify the model of a walking humanoid robot. In This simplified model the robot is represented by a 2-D cart-table of the same mass and at the height of the center of mass  $z_c$ .

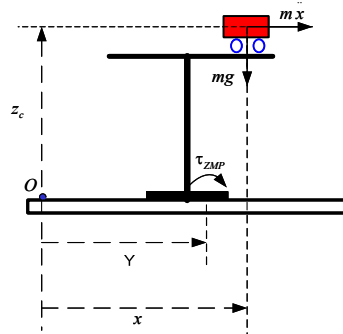


Fig.1.Cart Table Model

$m$ : The mass

$Y$  : The position of the (ZMP) COP

$x$  : The horizontal position of COM (center of mass)

$x'$  : The horizontal acceleration

$z_c$ : The altitude

$g$ : The norm of the gravity

This simplified model proves an extreme efficiency in the context of walking pattern generation. The equation of motion is represented by:

$$\tau_{ZMP} = mg(x - Y) - m \ddot{x} z_c = 0 \quad (1)$$

Where

$$Y = x - \frac{z_c}{g} \ddot{x} \quad (2)$$

Equation (2) obtained binds the ZMP position on an axis to the position and the acceleration of the COM on this same axis. Thus, a system of differential decoupled equations facilitating the resolution. Several methods can be used to solve this system of differential equations, analytically as in<sup>14</sup> or using the theory of predictive control as in<sup>13</sup>. The advantage of analytical solutions is their accuracy; in fact, these methods provide exact solutions to the system (2). However, their inclusion in a "real-time" system is more problematic. The predictive control method disadvantage is nevertheless the error inherent in the size of the prediction horizon. Given its ease of integration in the context of a "real-time" system, this method is chosen as a basis.

### 3. Pattern generation with preview control

Kajita et al<sup>1</sup> propose, for this pattern generation method, considering the jerk of the center of mass:

$$\frac{d \ddot{x}}{dt} = u_x \quad (3)$$

With the introduction of this new quantity, the equation of movement along  $x$  direction becomes :

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_x \\ Y &= \begin{bmatrix} 1 & 0 & -\frac{z_c}{g} \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \ddot{x} \end{bmatrix} \end{aligned} \quad (4)$$

The second equation is obtained from previous results in the cart-table model of the ZMP. This is a classic dynamic system with  $Y$  as known constant (predefined desired ZMP). Using the Taylor formula to the order three to discretize the equations with sampling period  $T$ , the following system is obtained:

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) \\ Y = Cx(k) \end{cases} \quad (5)$$

Where

$$\begin{aligned} x(k) &= \begin{bmatrix} x(kT) & \dot{x}(kT) & \ddot{x}(kT) \end{bmatrix} \\ u(k) &= u_k(kT) \\ Y(k) &= Y(kT) \end{aligned}$$

$$\left\{ \begin{array}{l} A = \begin{bmatrix} 1 & T & \frac{T^2}{2} \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}; B = \begin{bmatrix} \frac{T^3}{6} \\ \frac{T^2}{2} \\ 1 \end{bmatrix} \\ C = \begin{bmatrix} 1 & 0 & -\frac{z_c}{g} \end{bmatrix} \end{array} \right. \quad (6)$$

#### 4. Predictive control for the Cart-Table using GPC control

In this part of paper, the generalized predictive control law is used to synthesis the predictive PID. This method of control introduces the predictive output of process and the optimal control parameters for calculate the error between output signal and the set-point at a simple instant  $k$ . This obtained error can be put in the conventional form of PID controller<sup>3,7</sup>.

We start this study by considering the discrete-time state space representation of a SISO system, described by the equation (5).

In the first step we take the difference between  $x(k+1)$  and  $x(k)$ , we obtained the equation (7):

$$x(k+1) - x(k) = A(x(k) - x(k-1)) + B(u(k) - u(k-1)). \quad (7)$$

The following equations represent the difference between the state variables and the control variables given by:

$$\left\{ \begin{array}{l} \Delta x_d(k+1) = x_d(k+1) - x_d(k) \end{array} \right. \quad (8)$$

$$\left\{ \begin{array}{l} \Delta x_d(k) = x_d(k) - x_d(k-1) \end{array} \right. \quad (9)$$

$$\left\{ \begin{array}{l} \Delta u(k) = u(k) - u(k-1) \end{array} \right. \quad (10)$$

We connect  $\Delta x(k)$  to the output  $y(k)$  to obtain the integration effect and we chose the new augmented state vector  $X(k)$ .

$$X(k) = [\Delta x^T \quad y]^T \quad (11)$$

The new state space model with the embedded integrator is:

$$\left\{ \begin{array}{l} \begin{bmatrix} \Delta x(k+1) \\ y(k+1) \end{bmatrix} = \begin{bmatrix} A & 0^T \\ CA & 1 \end{bmatrix} \begin{bmatrix} \Delta x(k) \\ y(k) \end{bmatrix} + \begin{bmatrix} B \\ CB \end{bmatrix} \Delta u(k) \\ y(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta x(k) \\ y(k) \end{bmatrix} \end{array} \right. \quad (12)$$

With

$$\begin{cases} Ap = \begin{bmatrix} A & 0^T \\ CA & 1 \end{bmatrix} & Bp = \begin{bmatrix} B \\ CB \end{bmatrix} \\ Cp = [0 \quad 1] \end{cases}$$

In the rest of the paper, this augmented model will be used in the design of predictive control.

The prediction of the plant output with the future control variable is the important step in the predictive control design. Let the sampling instant  $k$ ; with  $k > 0$ .

$$\Delta u(k), \Delta u(k+1), \dots, \Delta u(k+nc-1) \quad (13)$$

The future vector of control movement is denoted by:

$$\Delta U = [\Delta u(k), \Delta u(k+1), \dots, \Delta u(k+nc-1)]^T \quad (14)$$

Where  $nc$  is called the control horizon, means that the variation inputs have zero in moments upper or equal to the control horizon. The control horizon can be  $nc \leq np$ .

The augmented state space, where  $Xp$  and  $Yp$  are the predicted input and output vectors respectively as:

$$\underbrace{\begin{bmatrix} x(k+1|k) \\ x(k+2|k) \\ \vdots \\ x(k+np|k) \end{bmatrix}}_{Xp} = \underbrace{\begin{bmatrix} A \\ A^2 \\ \vdots \\ A^{np} \end{bmatrix}}_H X(k) + \underbrace{\begin{bmatrix} B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{np-1}B & A^{np-2}B & \dots & A^{np-nc}B \end{bmatrix}}_{\Psi} \underbrace{\begin{bmatrix} \Delta u(k) \\ \Delta u(k+1) \\ \vdots \\ \Delta u(k+nc-1) \end{bmatrix}}_{\Delta U} \quad (15)$$

$$\underbrace{\begin{bmatrix} y(k+1|k) \\ y(k+2|k) \\ \vdots \\ y(k+np|k) \end{bmatrix}}_{Yp} = \underbrace{\begin{bmatrix} CA \\ CA^2 \\ \vdots \\ CA^{np} \end{bmatrix}}_F X(k) + \underbrace{\begin{bmatrix} CB & 0 & \dots & 0 \\ CAB & CB & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{np-1}B & CA^{np-2}B & \dots & CA^{np-nc}B \end{bmatrix}}_{\Phi} \underbrace{\begin{bmatrix} \Delta u(k) \\ \Delta u(k+1) \\ \vdots \\ \Delta u(k+nc-1) \end{bmatrix}}_{\Delta U} \quad (16)$$

## 5. Predictive PID approach based on GPC

### 5.1. Conventional PID structure

The discrete form of PID controller given by the following equation:

$$\Delta u(k) = [k_p \cdot e(k) + k_i \sum_{i=1}^k e(k) + k_d (e(k) - e(k-1))] \quad (17)$$

PID controllers could be written as <sup>7</sup>:

$$u(k) = \frac{\beta_0 + \beta_1 z^{-1} + \beta_2 z^{-2}}{1 - z^{-1}} e(k) \quad (18)$$

Where  $\beta_0 = k_p + k_i + k_d$ ,  $\beta_1 = -k_p - 2k_d$  and  $\beta_2 = k_d$ , and  $e(k)$  represents the error at sample k. Where the three gain of the conventional PID controller are  $k_p$ ,  $k_i$  and  $k_d$  are the proportional, integral and derivative gains respectively.

The velocity form of PID controller can be obtained with the control input increment at instant k:

$$\Delta u(k) = u(k) - u(k-1) = \beta_0 e(k) + \beta_1 e(k-1) + \beta_2 e(k-2) \quad (19)$$

The equation (19) can be written in matrices form as:

$$\Delta u(k) = K_{pid} E(k) = K_{pid} [y_r(k) - Y(k)] \quad (20)$$

With

$$E(k) = [e(k) \ e(k-1) \ e(k-2)]$$

Where

$$K_{pid} = [\beta_0 \ \beta_1 \ \beta_2] \quad (21)$$

## 5.2. Proposed predictive PID control

Consider the following quadratic cost function presented by equation (22), where the first principal part minimizes the errors between the reference signal and the output signal. And the second part minimizes the control effort. Propose  $y_r(k)$  the set-point signal at sample time k.

$$J_y = (y_r - Yp)^T (y_r - Yp) + \Delta U^T R \Delta U \quad (22)$$

The aim of the proposed predictive PID controller is to maintain the output signal provided as close as possible to the reference signal<sup>3</sup>.

The optimal parameter vector  $\Delta U$  can be obtained by substituting of equation of the new state space output give by (16) into (22),  $J_y$  is expressed as:

$$J_y = (y_r - FX(k))^T (y_r - FX(k)) - 2\Delta U^T \Phi^T (y_r - FX(k)) + \Delta U^T (\Phi^T \Phi + R) \Delta U \quad (23)$$

The necessary condition of the minimum J is obtained as:

$$\frac{\partial J_y}{\partial \Delta U} = 0$$

From the first derivative of the cost function  $J_y$ :

$$\frac{\partial J_y}{\partial \Delta U} = 2\Phi^T (y_r - FX(k)) + 2(\Phi^T \Phi + R) \Delta U = 0 \quad (24)$$

From which we find the optimal control vector solution as:

$$\Delta U = (\Phi^T \Phi + R)^{-1} \Phi^T (y_r - FX(k)) \quad (25)$$

With

$$e(k) = (y_r - FX(k)) \quad (26)$$

And

$$K_{GPCpid} = (\Phi^T \Phi + R)^{-1} \Phi^T \quad (27)$$

With the obtained results given by (27), we can conclude the equality of two gains:

$$K_{pid} = K_{GPCpid}$$

## 6. Simulation result

The simulation results obtained considers the following parameters:

Table 1. Simulation parameters.

| Parameters                |
|---------------------------|
| $T = 5[\text{ms}]$ ,      |
| $z_c = 0.814[\text{m}]$ , |
| $R = 1.0 \times 10^{-6}$  |
| $Q = 1.0 \times 10^{-6}$  |

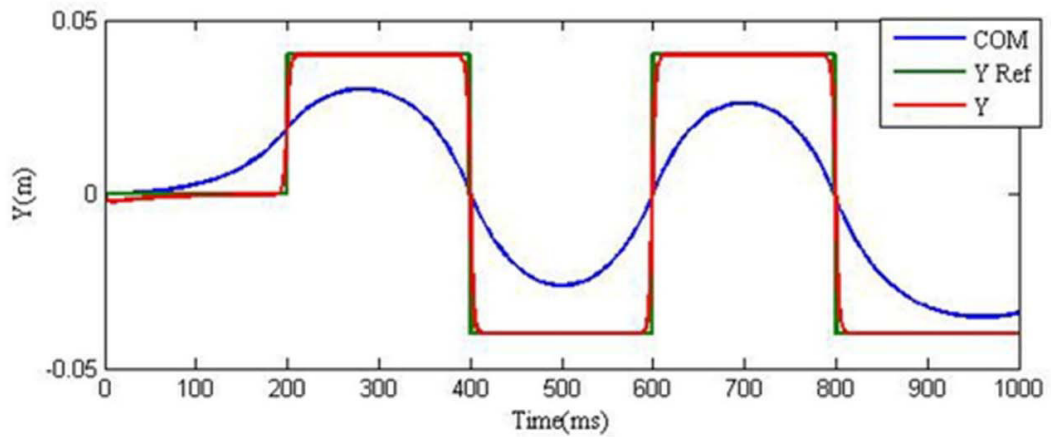


Fig. 2. Predictive state feedback using Predictive PID

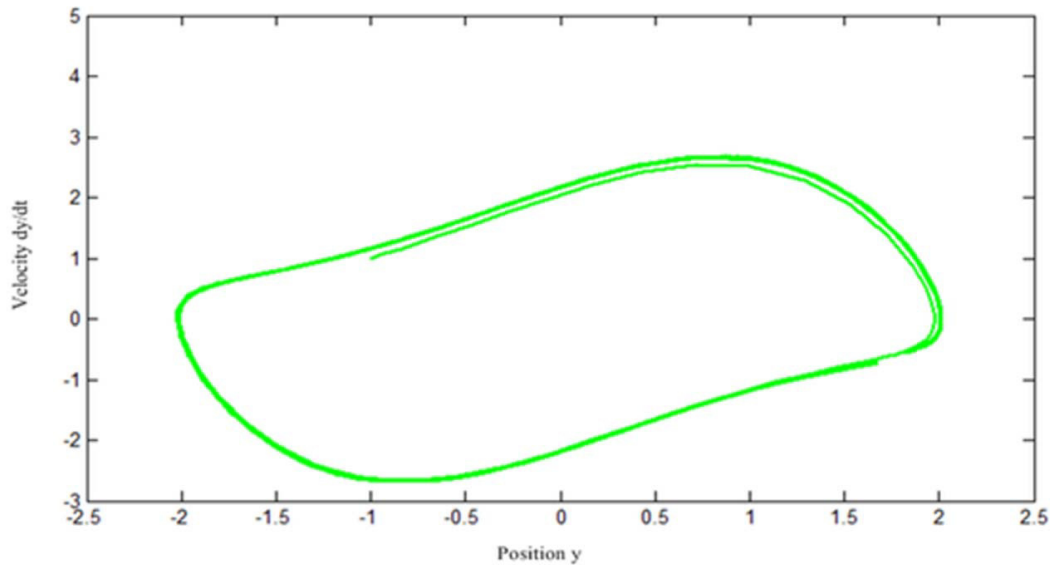


Fig. 3. Limit cycle of the Robot center of mass trajectory using Predictive PID

In the figure 2, we can see that the closed loop responses of the output system using the predictive PID is provided immediately and follow the set point, the center of mass give a stable limit cycle presents in figure 3. That can generate a stable gait cycle.

## 7. Conclusion

In this paper we apply and simulate a predictive controller for the control of a cart table model. Simulations confirm that the designed predictive PID controller have the capability to follow the reference and exhibits a fast response. PID adapted better to the real time control of the process due to the reduced number of calculation operations. The implementation of such a controller can be performed on any processor. However, the GPC algorithm needs powerful computation time and consequently requires fast processors dedicated to this type of calculation.

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